Multi-Variable Calculus
Assignment - 3

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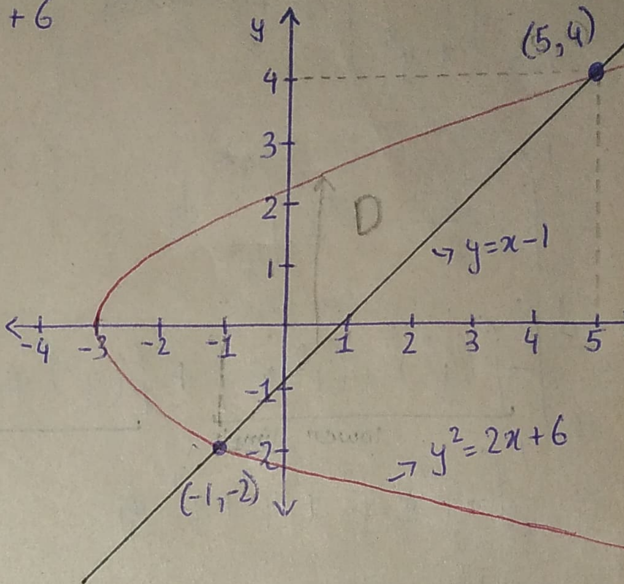
Q1) Evaluate $\iint_D xy \, dA$ where D is the region bounded by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$



Sol

$$\int_{y=-2}^{y=4} \int_{x=\frac{y^2-6}{2}}^{x=y+1} xy \, dx \, dy$$

$$= \int_{-2}^{4} \left. \frac{x^2 y}{2} \right|_{\frac{y^2-6}{2}}^{y+1} dy$$

$$= \int_{-2}^{4} \frac{(y+1)^2 y}{2} - \frac{y\left(\frac{y^2-6}{2}\right)^2}{2} \, dy$$

$$= \frac{1}{2} \int_{-2}^{4} y(y^2+2y+1) - y\left(\frac{y^4-12y^2+36}{4}\right) dy$$

$$= \frac{1}{2} \int_{-2}^{4} y^3 + 2y^2 + y - \frac{y^5}{4} + 3y^3 - 9y \, dy$$

$$= \frac{1}{2} \int_{-2}^{4} \frac{-y^5}{4} + 4y^3 + 2y^2 - 8y \, dy$$

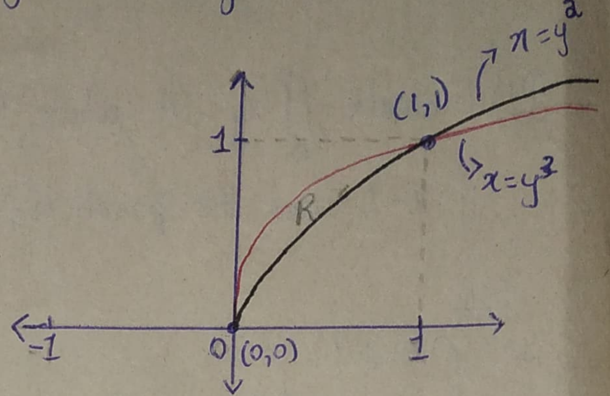
$$= \frac{1}{2} \left[\frac{-y^6}{24} + y^4 + \frac{2y^3}{3} - 4y^2 \Big|_{-2}^{4} \right]$$

$$= \frac{1}{2} \left[64 - (-8) \right]$$

$$= 36 \quad \text{Answer}$$

Q2) Find the volume of the solid under the surface $z = 2x + y^2$ and above the region bounded by $x = y^2$ and $x = y^3$

<u>Sol</u>



$$\int_{y=0}^{y=1} \int_{x=y^3}^{x=y^2} 2x + y^2 \, dx\,dy$$

$$= \int_0^1 \frac{2x^2}{2} + xy^2 \Big|_{y^3}^{y^2} dy$$

$$= \int_0^1 \underbrace{-\left[(y^3)^2 + (y^3)y^2\right]}_{\text{lower limit}} + \underbrace{\left[(y^2)^2 + (y^2)y^2\right]}_{\text{upper limit}} dy$$

$$= \int_0^1 -y^6 \pm y^5 + y^4 + y^4 \, dy$$

$$= \int_0^1 -y^6 \pm y^5 + 2y^4 \, dy$$

$$= -\frac{y^7}{7} \mp \frac{y^6}{6} + \frac{2y^5}{5} \Big|_0^1$$

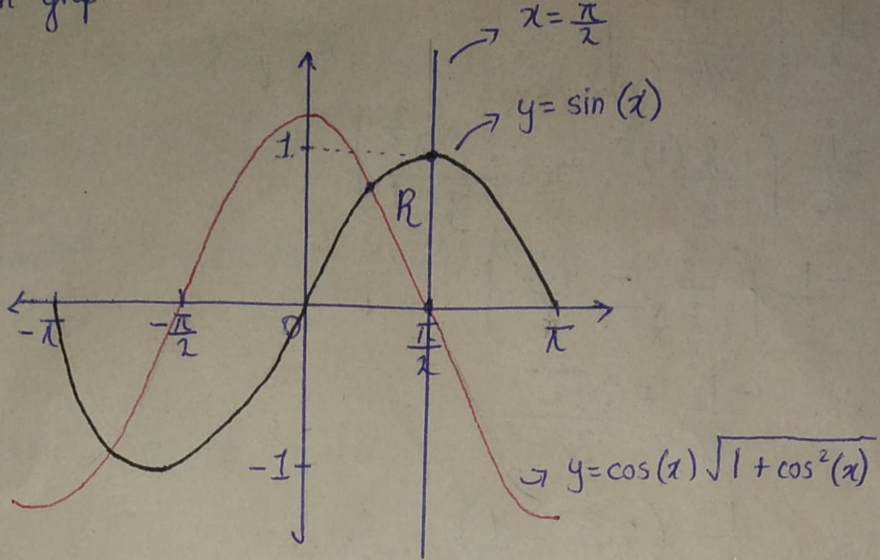
$$= \left(\frac{-1}{7} \mp \frac{1}{6} + \frac{2}{5}\right) - 0$$

$$= \frac{19}{120} \quad \text{Answer}$$

Q3) Evaluate the integral by reversing the order of integration

$$\int_{y=0}^{y=1} \int_{x=\sin^{-1}(y)}^{x=\pi/2} \cos x \sqrt{1+\cos^2 x} \, dx\,dy$$

First sketch graph



Reversing order

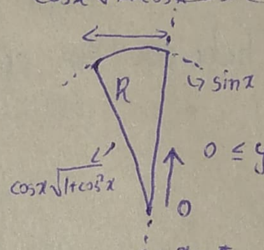
$$\int_{x=0}^{x=\frac{\pi}{2}} \int_{y=0}^{y=\sin(x)} \cos x \sqrt{1+\cos^2 x}\, dy\, dx$$

$$\int_0^{\frac{\pi}{2}} \cos x \sqrt{1+\cos^2 x} \cdot y \Big|_0^{\sin(x)} dx$$

$$\int_0^{\frac{\pi}{2}} \cos(x)\sin(x)\sqrt{1+\cos^2 x}\, dx$$

~~finding intersection~~

$\cancel{\cos x\sqrt{1+\cos^2 x} = \sin x}$



$0 \le y \le \sin x$

$0 \le x \le \frac{\pi}{2}$

$x = \sin^{-1}(y)$

$x = \sin^{-1}(1) = \frac{\pi}{2}$

$x = \sin^{-1}(y)$

$x = \sin(0)$

$x = 0$

using u substitution

$$u = 1+\cos^2 x$$

$$du = -2\cos x \sin x\, dx$$

$$dx = \frac{-1}{2\cos x \sin x} du$$

finding new limits

$x = 0 \rightarrow u = 1+\cos^2(0) = 2$ [~~upper~~ lower limit]

$x = \frac{\pi}{2} \rightarrow u = 1+\cos^2\left(\frac{\pi}{2}\right) = 1$ (~~lower~~ upper limit)

put it in integral

$$= \int_2^1 \cancel{\cos(x)}\,\cancel{\sin(x)}\sqrt{u} \cdot \frac{-1}{2\cancel{\sin x}\,\cancel{\cos x}}\, du$$

$$= \int_2^1 \frac{-\sqrt{u}}{2}\, du$$

$$= \frac{-1}{2}\int_2^1 u^{\frac{1}{2}}\, du$$

$$= \frac{-1}{2}\left[\frac{u^{\frac{3}{2}}}{3/2}\right|_2^1$$

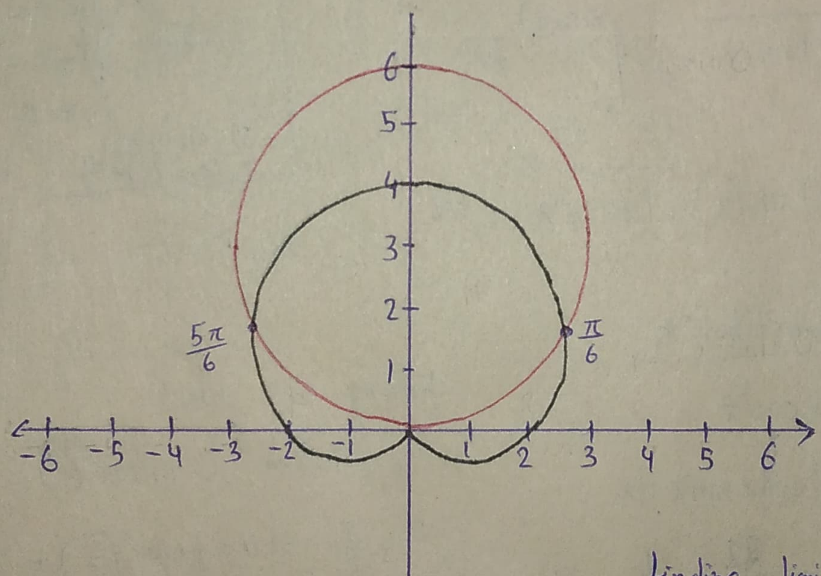
$$= \frac{-1}{2}\left[\frac{2(1)^{\frac{3}{2}}}{3} - \frac{2(2)^{\frac{3}{2}}}{3}\right]$$

$$= \frac{-1}{2}\left[\frac{2}{3} - \frac{4\sqrt{2}}{3}\right]$$

$$= \frac{-1}{3} + \frac{2\sqrt{2}}{3}$$

$$= \frac{1}{3}(2\sqrt{2}-1) \quad \text{Answer}$$

Q4) Use double integral to find the area of the region above the x-axis, between the cardioid $r = 2+2\sin\theta$ and circle $r = 6\sin\theta$



Solving

$$\int_{\theta=\frac{\pi}{6}}^{\theta=\frac{5\pi}{6}} \int_{r=2+2\sin\theta}^{r=6\sin\theta} r\,dr\,d\theta$$

$$\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \frac{r^2}{2}\Bigg|_{2+2\sin\theta}^{6\sin\theta} d\theta$$

finding limit

$6\sin\theta = 2+2\sin\theta$

$\sin\theta = \frac{2}{4}$

$\sin\theta = \frac{1}{2}$

$\theta = \underbrace{\frac{\pi}{6}}_{\boxed{\text{quadrant } 1}}$ and $\underbrace{\frac{5\pi}{6}}_{\boxed{\text{Quadrant } 2}}$

we get it

$\theta = \pi - \alpha$

$= \pi - \frac{\pi}{6} = \frac{5\pi}{6}$

$$\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (6\sin\theta)^2 - (2+2\sin\theta)^2 \, d\theta$$

$$\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} 36\sin^2\theta - 4 - 8\sin\theta - 4\sin^2\theta \, d\theta$$

$$\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} 32\sin^2\theta - 8\sin\theta - 4 \, d\theta$$ } took 4 common so $\frac{4}{2} = 2$

$$2\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \underbrace{8\sin^2\theta}_{1} \underbrace{- 2\sin\theta}_{2} \underbrace{- 1}_{3} \, d\theta$$

Solving each integral seperately for easiness.

Solving 1

$$\int 8\sin^2\theta \, d\theta$$

$$\cos 2\theta = 1 - 2\sin^2\theta$$
$$\sin^2\theta = \frac{1}{2} - \frac{\cos 2\theta}{2}$$

$$= 8\int \sin^2\theta \, d\theta$$
$$= 8\int \frac{1}{2} - \frac{\cos 2\theta}{2} \, d\theta$$
$$= 8\left[\frac{\theta}{2} - \frac{\sin 2\theta}{4}\right]$$
$$= 4\theta - 2\sin 2\theta$$

Solving 2

$$\int -2\sin\theta \, d\theta$$
$$= -2\int \sin\theta \, d\theta$$
$$= -2\left[-\cos\theta\right]$$
$$= 2\cos\theta$$

Solving 3

$$\int -1 \, d\theta$$
$$= -\theta$$

putting it all together again

$$2\left[4\theta - 2\sin 2\theta + 2\cos\theta - \theta \Big|_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \right]$$

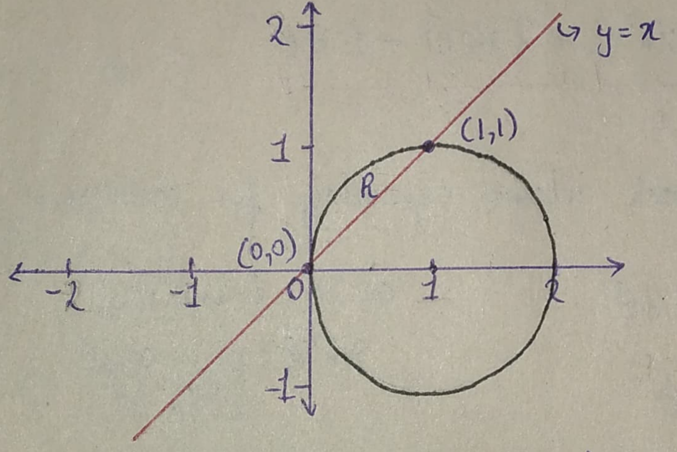
$$= 2\left[\frac{5}{2}\pi - \frac{\pi}{2}\right]$$

$$= 2\,[2\pi]$$

$$= 4\pi \quad \text{Answer}$$

Q5) Evaluate the iterated integral by converting to polar co-ordinates.

$\iint_D 2y\, dA$, where D is the region in the first quadrant bounded above by the circle $(x-1)^2+y^2=1$ and below by the line $y=x$.



Sol

$$\int_{\theta=\frac{\pi}{4}}^{\theta=\frac{\pi}{2}} \int_{r=0}^{r=2\cos\theta} 2(r\sin\theta)\, r\,dr\,d\theta$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{0}^{2\cos\theta} 2r^2 \sin\theta \; dr\,d\theta$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{2r^3 \sin\theta}{3} \Bigg|_0^{2\cos\theta} d\theta$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{2(8\cos^3\theta)\sin\theta}{3}\; d\theta$$

$$\frac{16}{3}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin\theta \cos^3\theta \; d\theta$$

finding limits

first quadrant and line $y = x$

$\therefore \quad \frac{\pi}{4} \le \theta \le \frac{\pi}{2}$

for r

$x = r\cos\theta, \quad y = r\sin\theta$

$(r\cos\theta - 1)^2 + (r\sin\theta)^2 = 1$

$r^2\cos^2\theta - 2r\cos\theta + 1 + r^2\sin^2\theta = 1$

$r^2(\cos^2\theta + \sin^2\theta) - 2r\cos\theta = 0$

$r^2 - 2r\cos\theta = 0$

$r^2 = 2r\cos\theta$

$r = 2\cos\theta$

$\therefore \quad 0 \le r \le 2\cos\theta$

Using u substitution

let $u = \cos\theta$

$du = -\sin\theta \, d\theta$

$d\theta = \frac{-1}{\sin\theta} du$

finding new limits

$u = \cos\left(\frac{\pi}{2}\right) = 0$

$u = \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

put it in integration again

$$\frac{16}{3} \int_{\frac{1}{\sqrt{2}}}^{0} u^3 \cancel{\sin\theta} \cdot \frac{-1}{\cancel{\sin\theta}} du$$

$$\frac{16}{3} \int_{\frac{1}{\sqrt{2}}}^{0} -u^3 \, du$$

$$\frac{-16}{3} \left[\frac{u^4}{4} \Big|_{\frac{1}{\sqrt{2}}}^{0} \right]$$

$$\frac{-16}{3} \left[0 - \frac{\left[\frac{1}{\sqrt{2}}\right]^4}{4} \right]$$

$$\frac{\cancel{-}\cancel{16}}{3} \left[\cancel{-} \frac{1}{\cancel{16}} \right]$$

$$= \frac{1}{3}$$ Answer

Q6) Evaluate the iterated integral

$$\int_{y=0}^{y=3} \int_{z=0}^{z=1} \int_{x=0}^{x=\sqrt{1-z^2}} z e^y \, dx\, dz\, dy$$

$$= \int_0^3 \int_0^1 x z e^y \Big|_0^{\sqrt{1-z^2}} dz\, dy$$

$$= \int_0^3 \int_0^1 (\sqrt{1-z^2}) z e^y - 0 \, dz\, dy$$

Using u substitution

let $u = 1 - z^2$

$du = -2z\,dz$

$dz = \dfrac{du}{-2z}$

finding new limits

$u = 1 - 1^2 = 0$ } upper limit

$u = 1 - 0^2 = 1$ } lower limit

put it back in integration

$$\int_0^3 \int_1^0 \sqrt{u}\, z e^y \cdot \frac{du}{-2z}\, dy$$

$$\int_0^3 \frac{-1}{2} \int_1^0 e^y \sqrt{u}\, du\, dy$$

$$= \int_0^3 \frac{-e^y}{2} \int_1^0 u^{\frac{1}{2}}\, du\, dy$$

$$= \int_0^3 \frac{-e^y}{2} \int \frac{2u^{\frac{3}{2}}}{3} \Big|_1^0\, dy$$

$$= \int_0^3 \frac{-e^y}{2} \left[-\frac{2}{3}\right] dy$$

$$= \int_0^3 + \frac{e^y}{3}\, dy$$

$$= \frac{1}{3} \int_0^3 e^y\, dy$$

$$= \frac{1}{3} \left[e^y \Big|_0^3 \right]$$

$$= \frac{1}{3} \left[e^3 - e^0\right]$$

$$= \frac{1}{3} \left[e^3 - 1\right]$$ Answer

Q7) Evaluate the integral $\int_C F \cdot dr$ where

$$F(x,y,z) = x\hat{i}\cancel{z} - z\hat{j} + y\hat{k}$$

and C is given by

$$r(t) = 2t\hat{i} + 3t\hat{j} - t^2\hat{k} \quad ; \quad -1 \le t \le 1$$

<u>Sol</u>

using formula 1 :-

$$\int_C \bar{F} \cdot d\bar{r} = \int_a^b \bar{F}(r(t)) \cdot r'(t)\, dt$$

finding $r'(t)$:

$$r'(t) = 2\hat{i} + 3\hat{j} - 2t\hat{k}$$

$$F(r(t)) = 2t\hat{i} - (-t^2)\hat{j} + 3t\hat{k}$$

finding integral;

$$= \int_{-1}^{1} (2t\hat{i} + t^2\hat{j} + 3t\hat{k}) \cdot (2\hat{i} + 3\hat{j} - 2t\hat{k})\, dt$$

$$= \int_{-1}^{1} 4t + 3t^2 - 6t^2\, dt$$

$$= \int_{-1}^{1} 4t - 3t^2\, dt$$

$$= \frac{4t^2}{2} - \frac{3t^3}{3} \Big|_{-1}^{1}$$

$$= 2t^2 - t^3 \Big|_{-1}^{1}$$

$$= (2(1) - 1^3) - (2(-1)^2 - (-1)^3)$$

$$= (2-1) - (2+1)$$

$$= -2 \quad \text{Answer}$$

Q8) Evaluate the line integral $\int_C xyz^2\, ds$, where C is the line segment from $(-1,5,0)$ to $(1,6,4)$

Sol

first make parametric equation

$$x(t) = x_0 + (x_1 - x_0)t = -1 + (1+1)t = -1+2t$$

$$y(t) = y_0 + (y_1 - y_0)t = 5 + (6-5)t = 5+t$$

$$z(t) = z_0 + (z_1 - z_0)t = 0 + (4-0)t = 4t$$

$$\therefore\ r(t) = (x(t) + y(t), z(t))$$

$$r(t) = (-1+2t)\hat{i} + (5+t)\hat{j} + 4t\hat{k}$$

using formula 2

$$\int_C F \cdot dr = \int_C F(x,y,z)\,ds = \int_a^b F(x(t), y(t), z(t)) \cdot \|r'(t)\|\,dt$$

finding $r'(t)$

$$r'(t) = 2\hat{i} + \hat{j} + 4\hat{k}$$

$$\|r'(t)\| = \sqrt{2^2 + 1^2 + 4^2} = \sqrt{21}$$

Put every thing in integral

$$= \int_0^1 (-1+2t)(5+t)(4t)^2 \cdot (\sqrt{21})\,dt$$

$$= \sqrt{21} \int_0^1 (-5 - t + 10t + 2t^2)(4t^2)\,dt$$

$$= \sqrt{21} \int_0^1 32t^4 + 144t^3 - 80t^2\,dt$$

$$= \sqrt{21} \left[\frac{32t^5}{5} + \frac{144t^4}{4} - \frac{80t^3}{3} \Big|_0^1 \right]$$

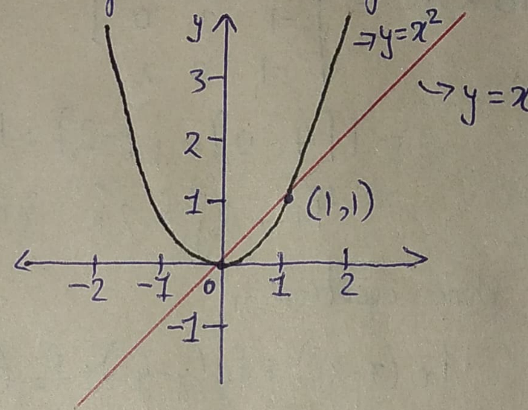
$$= \sqrt{21} \left[\frac{236}{15} - 0 \right]$$

$= \dfrac{236\sqrt{21}}{15}$ Answer

Q9) Use Green's Theorem to evaluate the line integral along the curve C:

$$\int_C (e^x + y^2)\, dx + (e^y + x^2)\, dy$$

where C is the boundary of the region between $y = x^2$ and $y = x$.



Solving RHS

$$\iint_R \frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\, dA$$

$$\int_{x=0}^{x=1} \int_{y=x^2}^{y=x} 2x - 2y \; dy\, dx$$

$$= \int_0^1 2xy - y^2 \Big|_{x^2}^{x} dx$$

$$= \int_0^1 2x^2 - x^2 - 2x^3 + x^4 \; dx$$

$$= \int_0^1 x^4 - 2x^3 + x^2 \; dx$$

$$= \frac{x^5}{5} - \frac{x^4}{2} + \frac{x^3}{3} \Big|_0^1$$

$$= \left[\frac{1}{5} - \frac{1}{2} + \frac{1}{3}\right] - [0]$$

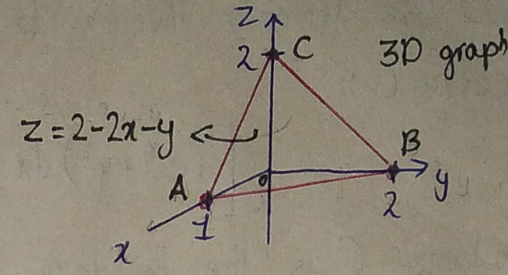
$$= \frac{1}{30}$$ Answer

Q10) Evaluate the surface integral $\iint_S xy \, dS$, when S is the triangular region with vertices $(1,0,0)$, $(0,2,0)$ and $(0,0,2)$.

Plane Equation

$\vec{AB} = [-1, 2, 0]$

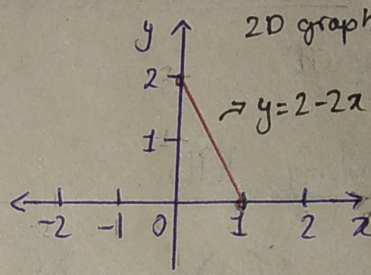
$\vec{AC} = [-1, 0, 2]$



$$\vec{AB} \times \vec{AC} = \begin{vmatrix} i & j & k \\ -1 & 2 & 0 \\ -1 & 0 & 2 \end{vmatrix}$$

$= i[4-0] - j[-2] + k[0+2]$

$= 4\hat{i} + 2\hat{j} + 2\hat{k}$ } normal vector



for plane equation

$f_x(x-x_1) + f_y(y-y_1) + f_z(z-z_1) = 0$

$4(x-1) + 2(y-0) + 2(z-0) = 0$

$4x - 4 + 2y + 2z = 0$

$z = +2 - 2x - y$ } eq of plane

finding limits (use 2D graph)

$0 \leq x \leq 1$

$0 \leq y \leq 2 - 2x$

$m = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{2-0}{0-1} = -2$

$y - y_1 = m(x - x_1)$

$y - 0 = -2(x-1)$

$y = 2 - 2x$

Parameterization

~~$r(x,y) = (x, y, 2-2x-y)$~~ let $z = g(x,y) = 2 - 2x - y$

$f(x, y, g(x,y)) = (x, y, +2-2x-y)$

$\dfrac{\partial g}{\partial x} = (1, 0, -2)$

$\dfrac{\partial g}{\partial y} = (0, 1, -1)$

finding $\left\| \frac{\partial g}{\partial x} \times \frac{\partial g}{\partial y} \right\|$

$$= \begin{vmatrix} i & j & k \\ 1 & 0 & -2 \\ 0 & 1 & -1 \end{vmatrix}$$

$$= i(0+2) - j(-1+0) + k(1)$$

$$= 2\hat{i} + \hat{j} + k$$

$$\left\| \frac{\partial g}{\partial x} \times \frac{\partial g}{\partial y} \right\| = \sqrt{(2)^2 + (1)^2 + (1)^2} = 6$$

Now do integration

$$\iint_S xy \, ds$$

$$\int_{x=0}^{x=1} \int_{y=0}^{y=2-2x} xy\sqrt{6} \, dy dx$$

$$\sqrt{6} \int_0^1 \int_0^{2-2x} xy \, dy \, dx$$

$$\sqrt{6} \int_0^1 \left. \frac{xy^2}{2} \right|_0^{2-2x} dx$$

$$\sqrt{6} \int_0^1 \frac{x(2-2x)^2}{2} - 0 \, dx$$

$$\frac{\sqrt{6}}{2} \int_0^1 x(4 - 8x + 4x^2) \, dx$$

$$\frac{\sqrt{6}}{2} \int_0^1 4x^3 - 8x^2 + 4x \, dx$$

$$\frac{\sqrt{6}}{2} \left[x^4 - \frac{8x^3}{3} + 2x^2 \Big|_0^1 \right]$$

$$= \frac{\sqrt{6}}{2}\left[\left(1-\frac{8}{3}+2\right)-0\right]$$

$$= \frac{\sqrt{6}}{2}\left(\frac{1}{3}\right)$$

$$= \frac{\sqrt{6}}{6}$$ Answer